Release Notes for

AT32F402_405 Firmware Library

V2.1.5-2026/06/02

1. CMSIS
   - system_at32f402_405.c
     - Updated the process of the System_init() to avoid potential clock failure during switching
   - system_at32f402_405.h
     - Added a new macro definition “SystemCoreClock” to prevent the occurrence of error when called by the third-party library
2. Middlewares
   - I2C
     - Corrected these macro definition typos: I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8; I2C_MEM_ADDR_WIDIH_16 is changed to I2C_MEM_ADDR_WIDTH_16
   - USB
     - Updated the I2C_MEM_ADDR_WIDTH_16 in host mode by adding a delay of 10ms after the SET_ADDRESS operation to ensure that the device address is taken into account before starting subsequent operation and configuration, to enhance compatibility
     - Updated the msc class by adding a step to handle the test_unit() and start_stop() to ensure better compatibility
     - Updated the msc class to address the problem of compiling warning issued in the uhost_process_handler() when the USBH_DEBUG_ENABLE is not activated
3. Drivers
   - TMR
     - Removed the tmr_output_channel_fast_set() and tmr_trgout2_enable()
   - PWC
     - __force_stores() is replaced with the __ISB() and the __DSB() to adapt to different compilers
   - CRM
     - Updated the processes of the crm_reset(), the crm_hick_sclk_frequency_select() and the crm_hick_sclk_frequency_select() to avoid potential clock failure during switching
   - EXINT
     - Added a new EXINT_TRIGGER_NONE_EDGE to adapt software trigger without edge requirements
4. Demos
   - I2C
     - Corrected these macro definition typos: I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8; I2C_MEM_ADDR_WIDIH_16 is changed to I2C_MEM_ADDR_WIDIH_16

- USB
  - Updated the description of the virtual_msc_iap demo to give a detailed information on how to clear interrupt because the previous one is not clear enough
5. Updated some notes

V2.1.4-2025/11/24

1. Drivers
   - TMR
     - Updated these functions: tmr_pwm_input_config(), tmr_encoder_mode_config() and tmr_output_channel_config()
       The channel must be disabled before configuring channel mode, otherwise it fails to switch between channel modes
2. Middlewares
   - USB
     - Updated "audio class" to change the audio sampling frequency request configuration, so as to solve host compatibility issue
3. Demos
   - ADC
     - Updated ADC initialization process in all demos to avoid errors occurring under extreme circumstances
     - Updated DMA-related demos to enable DMA circular mode during configuration to achieve circular trigger
     - Removed the software_trigger_repeat demo
   - USB
     - Updated Jump APP function in the virtual_msc_iap demo to add compiling non-optimization settings, so as to avoid Jump failure arising from optimization settings
   - Utilities
     - Updated the "qspi_algorithm_demo" to remove unnecessary configuration pr ocess and corrected the incorrect operation of using null pointer in code
   - AT32IDE and Eclipse projects
     - Removed "release" items and reserved "debug" items
   - IAP
     - Updated all IAP_Programmer.exe

V2.1.3-2025/06/25

4. CMSIS
   - device_support
     - Updated the "reduce_power_consumption()" function in *system_at32f402_405.c* and improved power consumption control process.
5. Drivers
   - CRM
     - Modified bit field "adc14rst" to "tmr14rst".

- CAN
  - Modified definitions of some register bit fields, with 14 filter banks only, and deleted redundant definitions.

6. Middlewares
   - USB
     - Updated keyboard class, added “send_state” member to the “keyboard_type” structure to record transmit status, and updated the “usb_keyboard_class_send_report” function for flow control processing in transmit to avoid data error.
7. Demos
   - USB
     - Adjusted the Deepsleep mode entry method (extra low-power mode) and configuration process in keyboard demo, and updated the delay of stabilization after wakeup.
     - Adjusted the Deepsleep mode entry method (extra low-power mode) and configuration process in mouse demo, and updated the delay of stabilization after wakeup.
     - Updated the value of “USB_FLASH_ADDR_OFFSET” in msc from 0x08005000 to 0x08008000 to avoid compilation range overflow in different environments.
   - templates
     - Updated the include path in eclipse_gcc to avoid compilation error caused by workspace_project_locations parameter error while importing multiple projects.
   - Board
     - Revised the writing of some parameter variables in board.c to avoid warning message in the new version of compiler.
   - Cortex-m4
     - Updated the delay judgment in systick_interrupt (modified “>” to be “>=”) to avoid the extra 1ms of IO toggling.
   - Keil project
     - Updated IROM size settings for some Keil projects.
8. Updated some notes.

V2.1.2-2025/01/20

1. Drivers
   - PWC
     - Updated LDO output level settings, and removed LDO configuration option "PWC_LDO_OUTPUT_1V0".
   - TMR
     - Updated "tmr_brk_filter_value_set()" function, and modified bit field "bkf" to "brkf".
2. Middlewares
   - USB
     - Updated "usbh_hch_in_handler()" function in host mode, and channels are disabled during interrupt transfer and synchronous transfer when NAK occurs.
3. Demos
   - IAP
     - Updated IAP_Programmer.exe.
4. Updated some notes.

V2.1.1-2024/12/13

1. CMSIS
   - DSP
     - Updated DSP source code library.
     - Updated *at32f402_405.h*, and corrected the encapsulated macro definitions (AT32F405Cx and AT32F402Cx).
     - Updated "system_core_clock_update()" function in *system_at32f402_405.c*, and fixed the error caused by calculation value overflow.
2. Drivers
   - CAN
     - Revised "can_transmit_cancel()" function writing to avoid cross-mailbox interference due to misoperation.
   - USB
     - Updated "usb_hch_halt()" function in host mode and improved channel halt processing flow.
     - Updated "usb_hc_enable()" function in host mode to ensure that its original state is cleared when enabling the corresponding channel.
     - Updated "usb_host_disable()" function in host mode, and updated process of disabling channels to avoid inadvertently opening the remaining channels.
   - SPI
     - Updated frequency calculation method in the "i2s_init()" function, and fixed the error caused by calculation value overflow.

- ERTC
  - Updated "ertc_wakeup_enable()" function, and corrected the use of parameters.
  - Updated "ertc_wait_update()" function, and standardized how this function is written to work across all series.
3. Middlewares
   - USB
     - Updated "usbd_reset_handler()" function, and changed endpoint status to NAK on USB RESET.
     - Added "usbd_ept_in_check_fifo()" function to check the current endpoint status and FIFO when transmitting data through USB.
     - Updated *mouse_class.c* and *custom_hid_class.c*, and added the process of checking whether the previous packet has been transferred completely when sending hid/mouse data.
     - Updated "usbd_set_configuration()" function in device mode, and modified the USB set configuration judgement logic.
4. Demos
   - I2C
     - Added memory_write demo.
     - Updated eeprom, and added multi-page write logic.
   - PWC
     - Updated the delay mechanism after waking up from Deepsleep mode in all PWC related demos to ensure stability.
   - SPI
     - Updated the structure and framework of all SPI demo codes, removed "usb_jtagpin_hardwarecs_dma" and added "fullduplex_dma".
   - QSPI
     - Updated QSPI interface configuration process (disabled cache) in "xip_port_read_write_sram".
   - Utilities
     - Added support for AT32IDE project in SLIB demo.
     - Updated bootloader jump address variables in usb_iap, and changed local variables to global variables to ensure that the jump address is not affected as much as possible.
5. Updated some notes.

V2.1.0-2024/08/08

1. Drivers
   - QSPI
     - Revised "qspi_flag_clear()" function writing, and fixed AC6 compilation problems.
   - USB
     - Updated "usb_hc_enable()" function in host mode, and fixed the enumeration issue of some devices.
     - Updated "usb_host_disable()" function in host mode, and fixed the issue that channels might not close properly in DMA mode.
     - Revised parameters in "usb_read_packet()" function, and fixed AC6 compilation problems.
2. Middlewares
   - USB
     - Updated "usbh_hch_in_handler()" function in host mode, without re-enabling channels during interrupt transfer.
     - Updated "uhost_process_handler()" function in host mode, and improved polling process.
3. Demos
   - Cortex-m4
     - Modified IROM base address and size settings of Keil project in cmsis_dsp.
   - I2C
     - Modified "GPIO_PULL_UP" to "GPIO_PULL_NONE" in I^2C GPIO initialization to enhance the adaptability of external circuits.
     - Modified DMA channel configuration sequence in "communication_dma" to prevent illegal pointer access.
   - SPI
     - Updated pin configurations in w25q_flash to better match the AT32-Comm-EV board.
   - USB
     - Modified write handling of *flash_fat16.c* file in virtual_msc_iap to enhance compatibility.
   - QSPI
     - Updated QSPI interface initialization and writing in all QSPI demos.
   - Utilities
     - Modified RAM size in IAR project link file of slib_demo/project_l1.
4. Updated some notes.

V2.0.9-2024/05/10

1. Drivers
   - USART
     - Updated "usart_init()" function and revised UART7/UART8 clock source.

V2.0.8-2024/04/30

1. CMSIS
   - device_support
     - Added "reduce_power_consumption()" function in system_at32f402_405.c for use in low power consumption control of at32f402.
2. Drivers
   - QSPI
     - Revised FIFO depth definition from "QSPI_DMA_FIFO_THOD_WORD32" to "QSPI_DMA_FIFO_THOD_WORD24".
   - SCFG
     - Adjusted the type of return value of "scfg_sram_operr_status_get()" function from "error_status" to "flag_status".
3. Demos
   - ALL at32f402 demos
     - Updated *at32f402_405_clock.c* file; call the "reduce_power_consumption()" function after the system clock is initialized.
   - SRAM
     - Updated "sram_parity_check" initialization process.
     - Updated "sram_parity_error_lock_enable" initialization process.

V2.0.7-2024/04/08

1. Drivers
   - QSPI
     - Added "qspi_auto_ispc_enable()" function for bus input clock phase matching.
2. Middlewares
   - USB
     - Revised STALL state assignment method in "usbh_hch_out_handler()" function.
3. Demos
   - QSPI
     - Updated all demos, and added "qspi_auto_ispc_enable()" function in initialization process.
   - USB_DEVICE
     - Updated USB low-power mode wakeup configuration and process to enhance system stability when entering/existing low-power mode.

V2.0.6-2024/03/01

1. Demos
    - QSPI
        - Added “run_in_qspi_flash” demo.
        - Added all *qspi_cmd_esmt32m.c* files, and removed unused “user_func_check_timeout()” function.
    - Utilities
        - Added qspi_algorithm_demo.

V2.0.5-2024/02/01

1. Demos
    - USB_DEVICE
        - Fixed the error caused by removing “PWC_REGULATOR_LOW_POWER” in mouse and keyboard demos.

V2.0.4-2024/01/26

1. CMSIS
    - device_support
        - Adjusted SRAM size in link files in gcc and IAR environment.
2. Drivers
    - PWC
        - Updated “pwc_voltage_regulate_set()” function and removed “PWC_REGULATOR_LOW_POWER” setting.
3. Middlewares
    - USB
        - Updated the type of “usbd_core_type” structure members to ensure 4-byte-alignment of “default_config” and “dev_config” addresses.
        - Updated “custom hid report” and “audio hid report” descriptors.
4. Demos
    - ALL Keil projects
        - Updated SRAM size configuration in all Keil projects.
5. Updated some notes.

V2.0.3-2024/01/05

1. Updated timer input_capture demo counting method.
2. Fixed HID low-speed identification problems in USB demo.
3. Updated systick initialization function in systick interrupt demo.
4. Added winusb demo.
5. Added ADC loop_transfer demo.
6. Updated the method to call “xx_interrupt_flag_get()” function in all demos.

V2.0.2-2023/10/28

1. Updated GPIO configuration in QSPI demo.
2. Updated USB drivers.
3. Updated some notes.

V2.0.1-2023/09/28

1. Updated the calculation process of the "system_core_clock" global variable.
2. Added dual OTG device and dual OTG host demo for AT32F405.
3. Revised SRAM size configuration in Keil project.
4. Updated USB keyboard and mouse descriptor drivers.

V2.0.0-2023/08/18

1. Initial release of AT32F402 and AT32F405 series firmware library.